# Reza Ahmadi

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**Prof. Riccardo Scandariato & The Selection Committee**
Institute for Software Security
Hamburg University of Technology (TUHH)
Hamburg, Germany

August 3, 2026

Dear Prof. Riccardo Scandariato and the Selection Committee,

My research interest lies in software security, especially software vulnerability localization. Building on a strong background from my Master's degree in Computer Engineering, and upon the recommendation of my supervisor Prof. Mariano Ceccato, I am applying to Hamburg University of Technology's PhD to conduct research with the top experts in this field. Earning a Ph.D. in Computer Science at TU Hamburg would allow me to pursue my research interests and give me new opportunities to contribute to the field.

Research in Automated Dynamic Analysis. I have been working in software security since my first semester in computer engineering at the University of Verona. The field focuses on automated techniques for identifying and assessing vulnerabilities in code. My work in the university is in automated API specification inference from network traffic, which seeks to unify two types of reasoning: static and dynamic. Efficient integration of automated dynamic analysis to extract structural knowledge from real-world software behavior is a major open problem that my thesis seeks to address. Our approach allows tools to execute long testing sequences that require many individual network requests, naturally guiding the analysis toward pinpointing software vulnerabilities.

Research in Machine Learning and Security. The two years of experience in modelling software architecture led me naturally to the booming field of machine learning and vulnerability localization. With a competitive research fellowship, I undertook a new project on designing Physics-Informed Neural Networks (PINNs) and using PyTorch with our academic collaborators at the INTELLIGO Labs. By the end of the semester, I was fluent in programming with Java, Python, and C++, the various machine learning methods, API security, and MITM/SSL network interception. Armed with these research tools, I dedicated my thesis to automated dynamic analysis, with my co-authors, culminating in a manuscript submitted to SCAM 2026.

In addition to research, I strive to be a reliable creator of software in any industry I can. To this end I have held many engineering and development roles. I served as a Software Developer for Bale Messenger, Basalam, and Reply, which are early major applications with over a million users from diverse backgrounds. My past experiences developing and optimizing my software have made me strong at communicating complex architectures to a variety of environments and codebases at scale.

The Institute of Software Security is a top destination for my graduate research because of its strong expertise in software vulnerability localization. Specifically, I would be excited to work with Prof. Riccardo Scandariato. Continuing my proposed work under his guidance would provide me a strong environment for pursuing my interests and advancing my research. Finally, for the committee and faculty: Thank you very much for your time and consideration, I look forward to hearing from you soon.

Sincerely,

**Reza Ahmadi**